give you an approximate idea of the different concepts employed in physics these days in order to illustrate the structure of the atom, in some degree at least. The reason why no such visual representation can ever manage to cover simultaneously every characteristic of atomic structure, has been discussed already.

III. THE PERIODIC SYSTEM OF ELEMENTS

We come now, finally, to the question of the relationship between the chemical properties of the elements and the structure of the extranuclear parts of their atoms, the number of planetary electrons, and thus, ultimately, the nuclear charges of these atoms. We are indebted to Bohr's theory for our understanding of this interrelationship, and we can obtain the most satisfactory general idea of it by arranging the elements according to the magnitude of their respective nuclear charges, in other words, according to their atomic numbers, which are always identical with the number of elementary quanta of positive electricity on their respective nuclei. (Table II, at the end of this book.) Thus, we start with hydrogen (1), continue with helium (2), and proceed in this manner until we reach curium (96). It has been known to chemists for about a hundred years that the chemical properties of the elements repeat themselves when they are arranged in the order of their atomic numbers. If we break off the series at the end of each such period and start a new line; we get the well known *periodic system of elements* of Newlands, Mendelejeff and Mayer (Table III, at the end of this book). According to Bohr, this periodicity of the elements can be explained on the ground of atomic theory as follows:

According to a principle formulated by Pauli, one orbit cannot be occupied by more than one electron. There is room for one electron only in any one orbit (or in any one stationary vibration with a specific quantum number). This principle may be cited here without proof. So, if an atom has several electrons, the electrons after the first one will be located in ever farther outlying orbits. In considering such problems, it is always advisable to refrain from imagining a completed atom and to proceed, instead, outward from the bare nucleus, and to imagine that one electron is added after another, outwards from the centre, until finally the number of electrons characteristic of the element is completed.

If we continue with the successive addition of individual electrons according to this structural principle, we find that after the incorporation of a specific number of electrons, these electrons constitute, in an important sense, a complete system, the addition of further electrons begins, as it were, a new structure at a considerably greater distance from the nucleus. The extranuclear